

Ze Zheng Guan

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EDUCATION

- Southern University of Science and Technology, Shenzhen, China 2023 – 2026
 - Master’s student in Intelligent Manufacturing and Robotics
 - Advisor: Prof. Seungwoo Je
 - Research focus: Haptics, Virtual Reality, and Interaction Design
- Sanjiang University, Nanjing, China 2018 – 2022
 - Bachelor of Engineering in Automation

PUBLICATIONS

- Hold the Line: Restoring Artistic Expression in VR for People with Parkinson’s
Qianyan Zou, Zhuang Chang, **Ze Zheng Guan**, Zirui Xiao, Huidong Bai, Mark Billinghurst, Xueliang Li, Seungwoo Je
Published in *International Journal of Human-Computer Studies*
- PinCity: Multiplayer AR System with Shape-Changing Displays for Dynamic Urban Design
Tianze Xie, Hanzhong Luo, Zhicheng Wang, **Ze Zheng Guan**, Seungwoo Je
Published in *Proceedings of SIGGRAPH Asia 2025 Emerging Technologies*
- HaPet: A Desktop-Level Hemispherical Encountered-Type Haptic Display Providing Visuo-Haptic Feedback for Virtual Pet Interaction in VR
Ze Zheng Guan, Junyu Chen, Xuesong Zhang, Seungwoo Je
Published in *Proceedings of the Chinese CHI Conference 2024*
- ThermoShift: Enhancing VR Immersion with Thermal Motion Simulation
Ke Ding, Ruxiao Chen, **Ze Zheng Guan**, Seungwoo Je
Published in *Proceedings of the Chinese CHI Conference 2024*
- ShapeBall: A Shape-changing Display for Bimanual Haptic Interaction in Virtual Reality [under review].
- Resized Bimanual Grasping in VR: Characterizing Object Size Discrimination in Bimanual Interaction [under review].
- Flat2Curve: Interacting with Curved Displays in Virtual Reality using a Flat Physical Pad [under review].

PATENTS

- Spherical Haptic Device, Driving Method, and Virtual Reality System
Seungwoo Je, **Ze Zheng Guan**
CN120653124B, 2025-11-25
- Landscape Generation System
Seungwoo Je, Tianze Xie, Yilong Lin, Yuxin Ma, Kaihan Wang, **Ze Zheng Guan**, Yingjie Chang
US20260065809A1, Pending
- Shape Variable Display Array
Seungwoo Je, Tianze Xie, Yilong Lin, Yuxin Ma, Kaihan Wang, **Ze Zheng Guan**, Yingjie Chang
US12524024B1, 2026-01-13
- Landscape Generation System
Seungwoo Je, Tianze Xie, Yilong Lin, Yuxin Ma, Kaihan Wang, **Ze Zheng Guan**, Yingjie Chang
CN118815251A, 2024-10-22
- Landscape Generation System
Seungwoo Je, Tianze Xie, Yilong Lin, Yuxin Ma, Kaihan Wang, **Ze Zheng Guan**, Yingjie Chang
CN223410557U, 2024-10-03

- Landscape Generation System
Seungwoo Je, Tianze Xie, Yilong Lin, Yuxin Ma, Kaihan Wang, **Ze Zheng Guan** Yingjie Chang
CN118704836A, 2024-09-27

PROJECTS

- A Pin-Array Display for Iterative Exploration of a Community's Shared Vision 2025
In collaboration with the City Science Group, MIT Media Lab.
- Pin-Array-Based Landscape Generation System 2024
Commercial project, collaborating with DCBOLT and CBRE.
- TaBot: AI-Empowered Robotic Table for Balanced Conversations 2024
Exhibition project, in collaboration with X-Group at SUSTech.
- Intelligent Poetry and Painting Creation System Based on Gesture Interaction 2024
Research project, in collaboration with X-Group at SUSTech.

TALK & EXHIBITION

TALK:

- ACM Chinese CHI Conference 2024
- DS Student Research Sharing Seminar 2024

EXHIBITION:

- The 10th Bi-City Biennale of Urbanism/Architecture (UABB, Shenzhen) 2025
- The 21st China (Shenzhen) International Cultural Industry Expo and Trade Fair 2025
- University of Melbourne's 2024 Edible Urban Futures Winter Intensive 2025
- International Design Week 2024
- China Hi-Tech Fair Design Week 2023
- Maker Faire Shenzhen 2023

TEACHING ASSISTANT

- Cross-Time-Zone Collaborative Design, SUSTech — Prof. Enza Migliore Fall 2024
- Design Ethics, SUSTech — Prof. Thomas Fischer Spring 2024
- Additive Manufacturing, SUSTech — Prof. Seungwoo Je Fall 2023

TECHNICAL & RESEARCH SKILLS

- Haptic device prototyping, embedded system development, PCB design
- Study design and data analysis (user studies, psychophysics, HCI evaluation methods)